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## Questions

**Q1.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

A cylinder of aluminium has a weight of 35.0 N and a volume of $1.32 \times 10^{-3}$ $m^3$ .

Which of the following calculations gives the density of aluminium in kg $m^{-3}$ ?

☐ **A** $\dfrac{9.81 \times 1.32 \times 10^{-3}}{35.0}$

☐ **B** $\dfrac{1.32 \times 10^{-3}}{35.0 \times 9.81}$

☐ **C** $\dfrac{35.0}{9.81 \times 1.32 \times 10^{-3}}$

☐ **D** $\dfrac{35.0 \times 9.81}{1.32 \times 10^{-3}}$

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**(Total for question = 1 mark)**

**Q2.**

The photograph shows a floating object called a buoy. A long chain attaches the buoy to a very large mass at the bottom of the sea so that the buoy remains stationary.

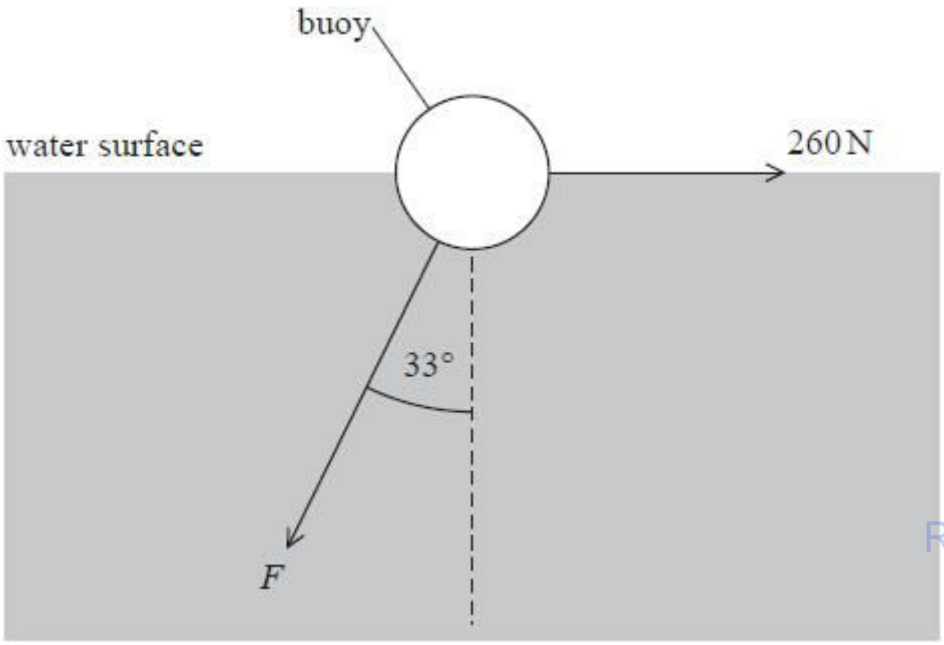


(Source: © EThamPhoto/Alamy Stock Photo)

Water flowing past the buoy causes a horizontal force of 260 N on the buoy.
The chain exerts a force *F* on the buoy at an angle of 33° to the vertical as shown.

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(a) (i) Show that *F* is about 500 N.

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(ii) The buoy floats due to the upthrust from the water.
The weight of water displaced by the buoy is 2.9 kN.
Determine the weight of the buoy.

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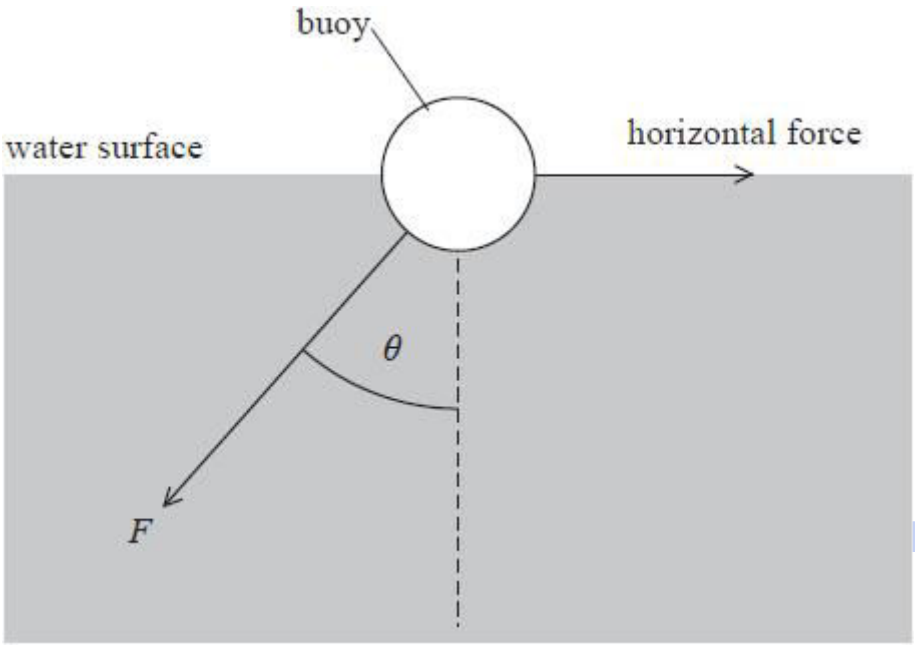
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Weight of buoy = .....

(b) The speed of the water flowing past the buoy increases, so the horizontal force increases. Assume that the weight of water displaced by the buoy does not change.

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Explain how $F$ and $\theta$ change when the horizontal force increases.

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**(Total for question = 8 marks)**

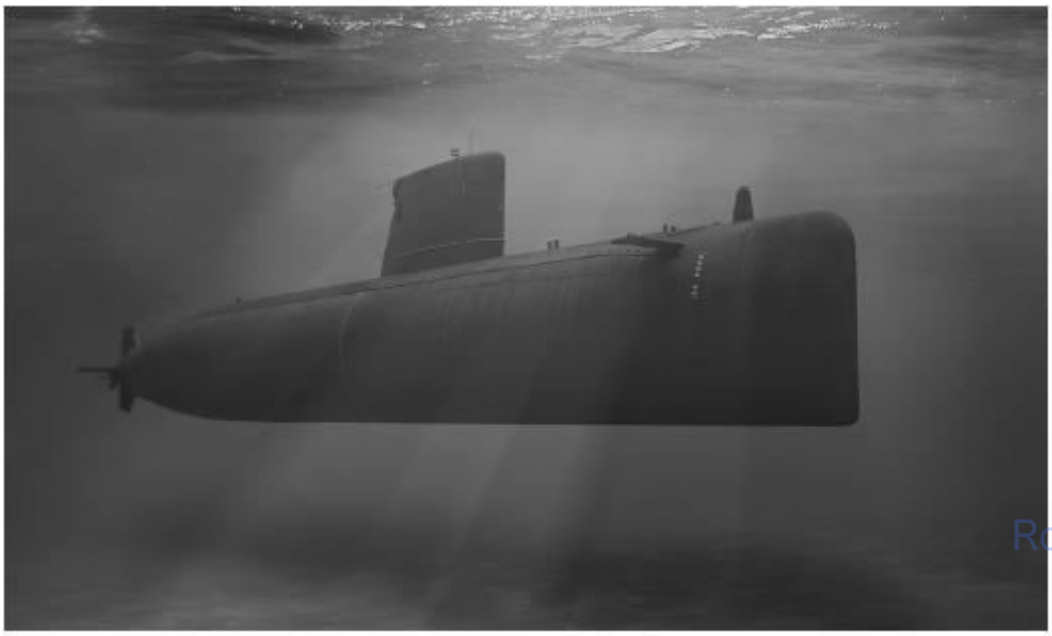
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**Q3.**

The photograph shows a submarine below the surface of the sea.

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(Source: noraismail/Shutterstock)

(a) The submarine has a volume of $5.83 \times 10^3$ $m^3$ .

The submarine is stationary in a region of the sea where the density of the seawater is $1.03 \times 10^3$ kg $m^{-3}$ .

(i) Calculate the upthrust exerted on the submarine by the seawater.

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Upthrust = .....

(ii) Explain why the mass of the submarine must be $6.0 \times 10^6$ kg.

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(b) The submarine now moves into a region of the sea where the water is less salty, and the density of the water reduces to $1.01 \times 10^3$ kg $m^{-3}$ .

(i) Explain what would happen to the submarine as it moves into this region of lower density seawater.

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(ii) The submarine alters its weight by pumping water in or out of its internal tanks. Determine the mass of water that the submarine should pump, in or out of its tanks, to maintain its depth below the surface of the sea.

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Mass of water = .....

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**(Total for question = 9 marks)**

**Q4.**

The photograph shows a fireboat used to put out fires on ships at sea. A pump, fixed to the boat, pumps water from the sea. The seawater is projected at high speed out of a pipe connected to the pump.



(Source: © Konrad Zelazowski/age fotostock/Superstock)

(a) The mass of seawater pumped each second is 300 kg. The pipe has a diameter of 10.0 cm.

density of seawater = 1030 kg $m^{-3}$

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(i) Show that the speed at which the seawater is projected from the pipe is about 37 m $s^{-1}$.

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(ii) Determine the rate at which the momentum of the seawater is changed by the pump.
You may assume that the seawater is initially stationary.

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Rate of change of momentum = .....

(b) Projecting water from the pipe causes a force to be exerted on the pump.

Explain the direction of the force on the pump.

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(c) Initially the pump is turned off and the fireboat moves forwards through the water at a constant speed. The boat's engine provides a constant forward force.

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When the pump is turned on, water is projected forwards and the fireboat slows to a lower constant speed.
Explain why the boat now has a lower constant speed.
Your answer should refer to all the horizontal forces on the boat.

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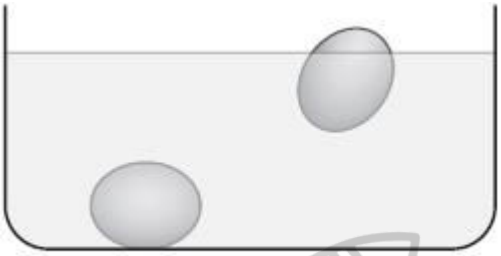
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**(Total for question = 11 marks)**

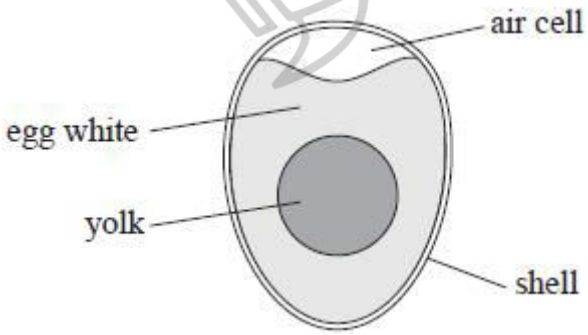
**Q5.**

* The approximate age of an egg can be determined by placing it in a bowl of water. Two eggs of different ages are placed in water and come to rest as shown.

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The internal structure of an egg is shown below.




A student searched on the internet to find the reason why old eggs float.

He found the following statements on different websites.

Statement 1

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| Old eggs float because as the egg ages it starts to decompose. As it decomposes gases are produced that escape through the eggshell. |

Statement 2

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| As the egg ages, air enters the egg through the eggshell and increases the size of the air cell. The larger air cell acts as a flotation device and hence old eggs float. |

Assess which of these two statements is correct.

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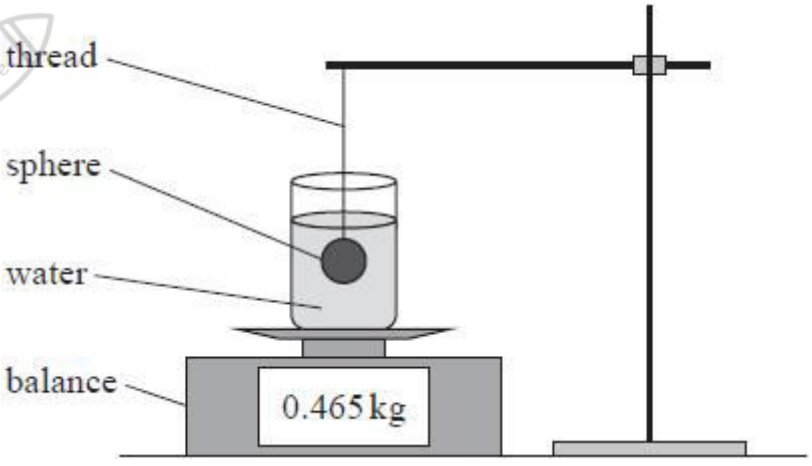
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**(Total for question = 6 marks)**

**Q6.**

A student used the apparatus shown to investigate the forces acting on a uniform, solid sphere placed in water.




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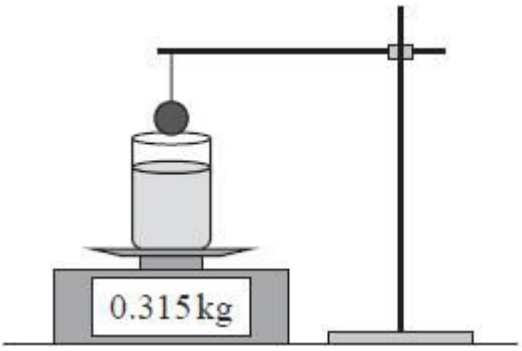
(a) Draw a free-body force diagram for the sphere when it is suspended in water as shown.

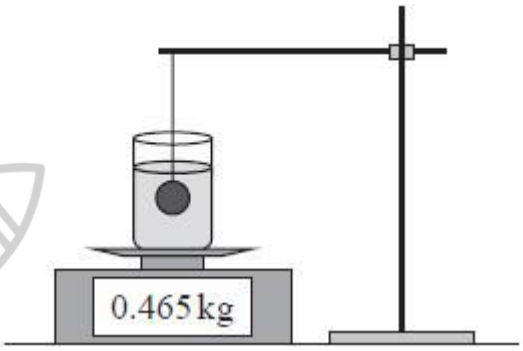
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(b) As the sphere was lowered into the water, the reading on the balance increased from 315 g to 465 g as shown.

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(i) Explain the increase in the reading on the balance. Your answer should include a reference to Newton's third law.

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(ii) Determine the mass of the sphere.
density of water = 1000 kg $m^{-3}$
density of sphere = 2000 kg $m^{-3}$

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Mass of sphere = .....

(iii) The water was replaced with the same mass of oil and the experiment repeated.
The density of the oil is less than the density of water.

Explain what happens to the final reading on the balance.

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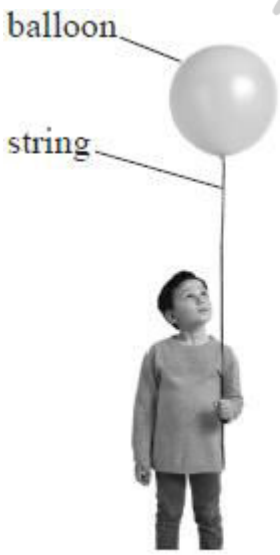
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**(Total for question = 12 marks)**

**Q7.**

A child is holding a balloon by a string, as shown. The balloon contains helium gas.
Helium gas is less dense than air.




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(Source: © pinstock/Getty Images)

(a) The child releases the balloon by letting go of the string.

Calculate the resultant force on the balloon immediately after it is released.
mass of balloon and string = $9.20 \times 10^{-3}$ kg
volume of balloon and string = $5.00 \times 10^{-2}$ $m^3$
density of air = 1.20 kg $m^{-3}$

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Resultant force = .....

*(b) Immediately after the balloon is released, it begins to move upwards.

The density of the air remains constant and there is no wind.
Explain how the forces acting on the balloon affect its motion as it moves upwards from the moment it is released.

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**(Total for question = 10 marks)**

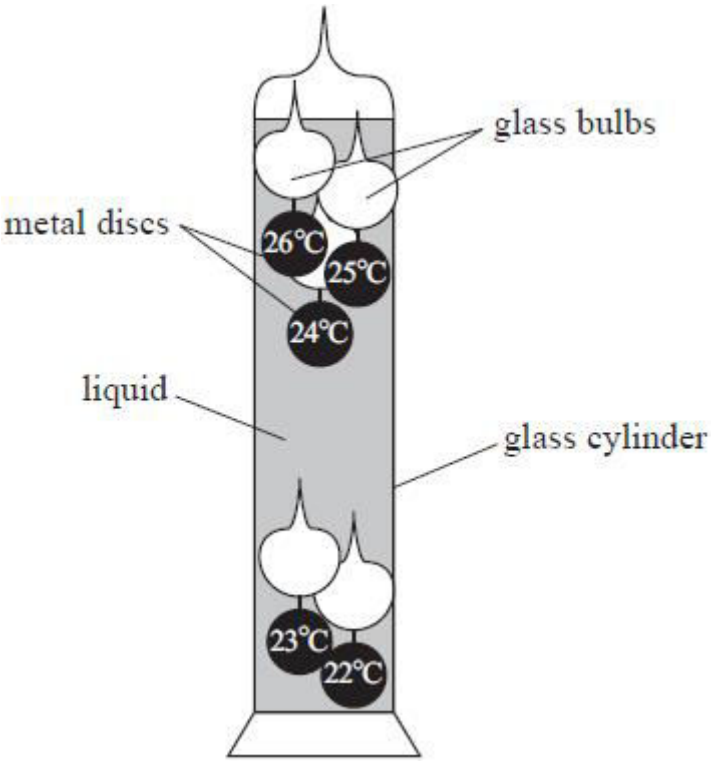
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**Q8.**

A Galilean thermometer consists of a closed glass cylinder containing a liquid. In the liquid there are several identical sealed glass bulbs, as shown. Attached to each bulb is a metal disc labelled with a temperature. Each disc has a different mass.

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As the temperature increases, the density of the liquid decreases. This can cause the bulbs to move within the liquid.

Explain why a particular bulb will float until the temperature of the liquid exceeds a certain value.

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**(Total for question = 6 marks)**

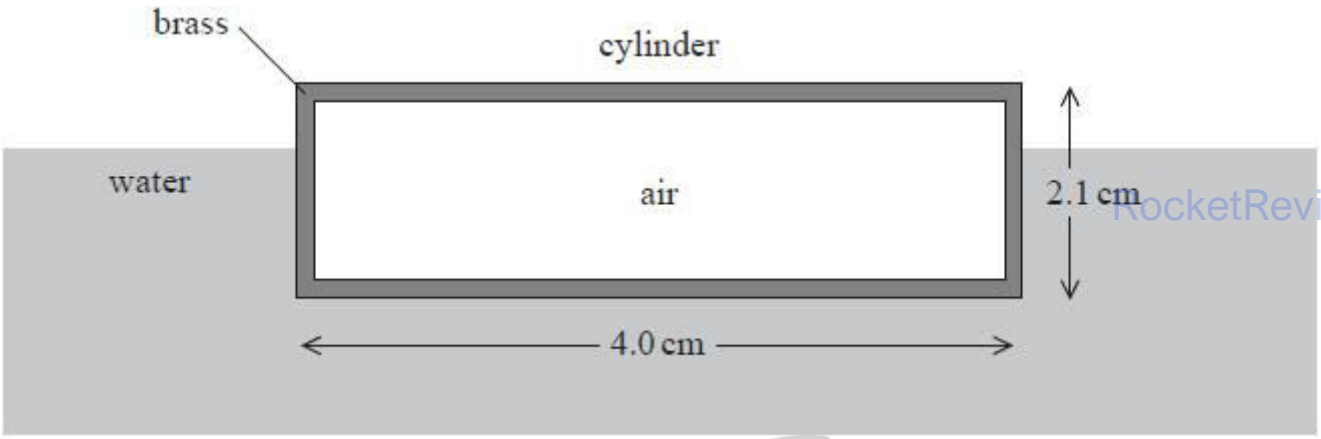
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**Q9.**

A hollow brass cylinder with closed ends is floating on the surface of water.

The cylinder has a length of 4.0 cm and an external diameter of 2.1 cm as shown.




63% of the volume of the cylinder is submerged. The cylinder contains negligible weight of air.

(a) Explain why the brass cylinder floats.

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(b) The density of water is $1.0 \times 10^3$ kg $m^{-3}$

(i) Show that the mass of the cylinder is about $9 \times 10^{-3}$ kg.

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(ii) Deduce whether an identical hollow cylinder made of gold would also float.
Assume that the volume of gold is the same as the volume of brass.
density of gold = $19.3 \times 10^3$ kg $m^{-3}$
density of brass = $8.7 \times 10^3$ kg $m^{-3}$

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**(Total for question = 10 marks)**

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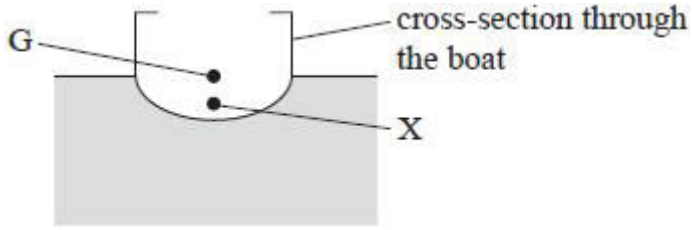
**Q10.**

For a boat to maintain a constant upright position in the water, the weight *W* and upthrust *U* should have the same line of action.

*W* acts through G, the centre of gravity of the boat.

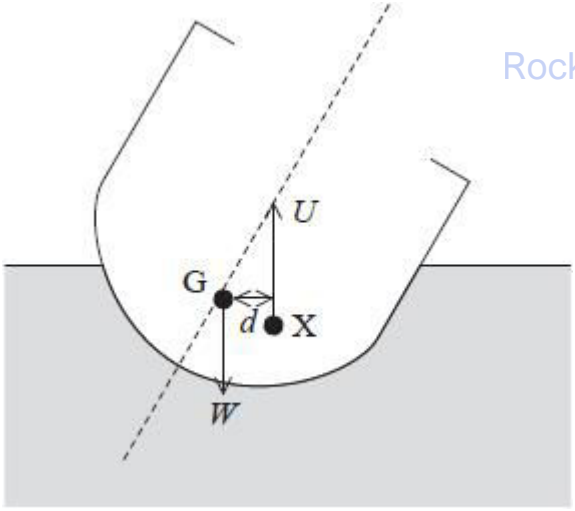
*U* acts through X, the centre of gravity of the volume of water displaced by the boat.

G and X are shown on the diagram.




(a) A sudden gust of wind applies a force to the side of the boat, causing it to tilt.

The lines of action of *W* and *U* move apart a distance *d*, as shown below.




(i) Explain the effect of the moment *Ud*.

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(ii) Explain what would happen to the boat in windy weather if the centre of gravity G were higher.

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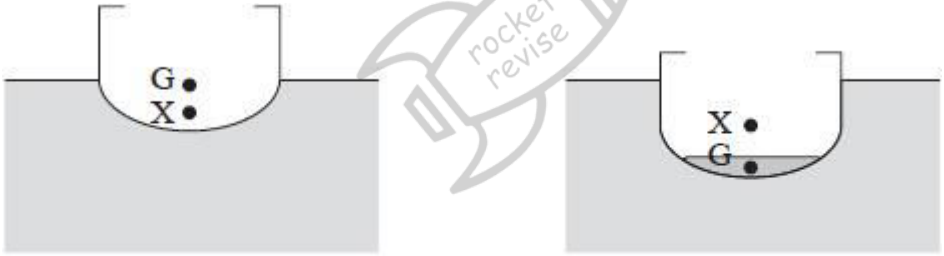
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(b) Some boats use a ballast tank, which is a refillable tank of water in the base of the boat, to improve stability.




Ballast tank empty Ballast tank full

(i) Explain why the position of X for the ship when it has a full ballast tank is lower than when the ship has an empty ballast tank.

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(ii) Explain a disadvantage of using a full ballast tank when the boat is moving through the water.

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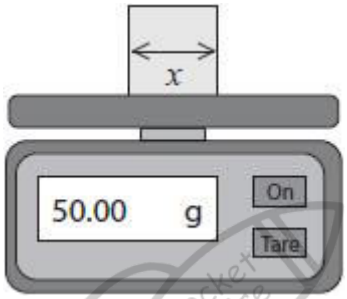
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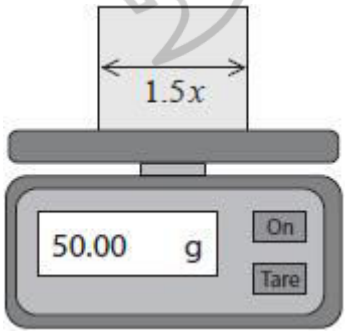
**Q11.**

A student used a balance to measure the mass of a small cube with sides of length $x$.

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The student also measured the mass of a larger cube with sides of length 1.5 $x$.




Which of the following is the density $\rho_L$ of the larger cube in terms of the density $\rho_S$ of the smaller cube?

- ☐ **A** $\rho_L = 3.4\ \rho_S$
- ☐ **B** $\rho_L = 1.5\ \rho_S$
- ☐ **C** $\rho_L = 0.67\ \rho_S$
- ☐ **D** $\rho_L = 0.30\ \rho_S$

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**(Total for question = 1 mark)**

**Q12.**

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**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

A sphere of weight 2.5 N floats in water with ½ of its volume beneath the surface.

A force *F* is applied to the sphere, completely immersing it in the water as shown.




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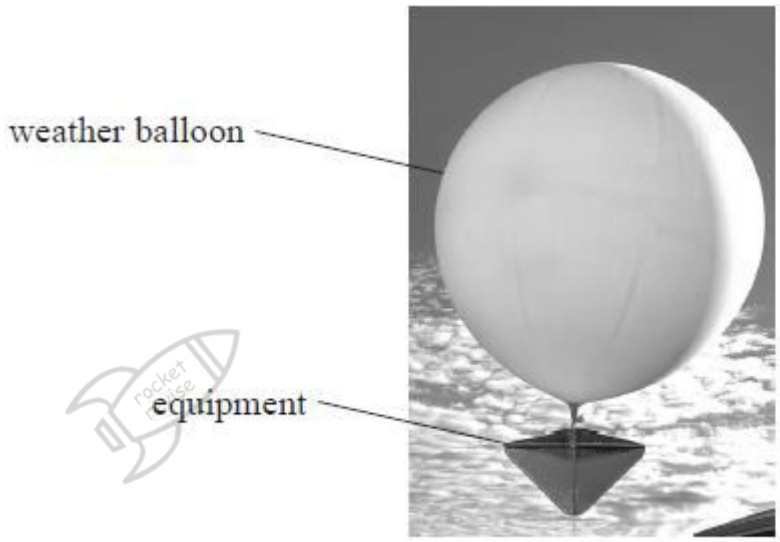
Which of the following is the minimum value of *F*?

- ☒ **A** 2 × 2.5 N
- ☒ **B** 1 × 2.5 N
- ☒ **C** ½ × 2.5 N
- ☒ **D** ¼ × 2.5 N

**(Total for question = 1 mark)**

**Q13.**

Scientists use weather balloons to carry equipment into the upper atmosphere to monitor weather conditions. The photograph shows a weather balloon.




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(Source: © Edward Haylan/Shutterstock)

Upthrust acts on the balloon. The balloon is released and accelerates upwards.

(a) State what is meant by upthrust.

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(b) (i) When the balloon is released, the initial upward acceleration is 4.80 m $s^{-2}$.

volume of balloon and equipment = 1.63 $m^3$
density of air at ground level = 1.23 kg $m^{-3}$

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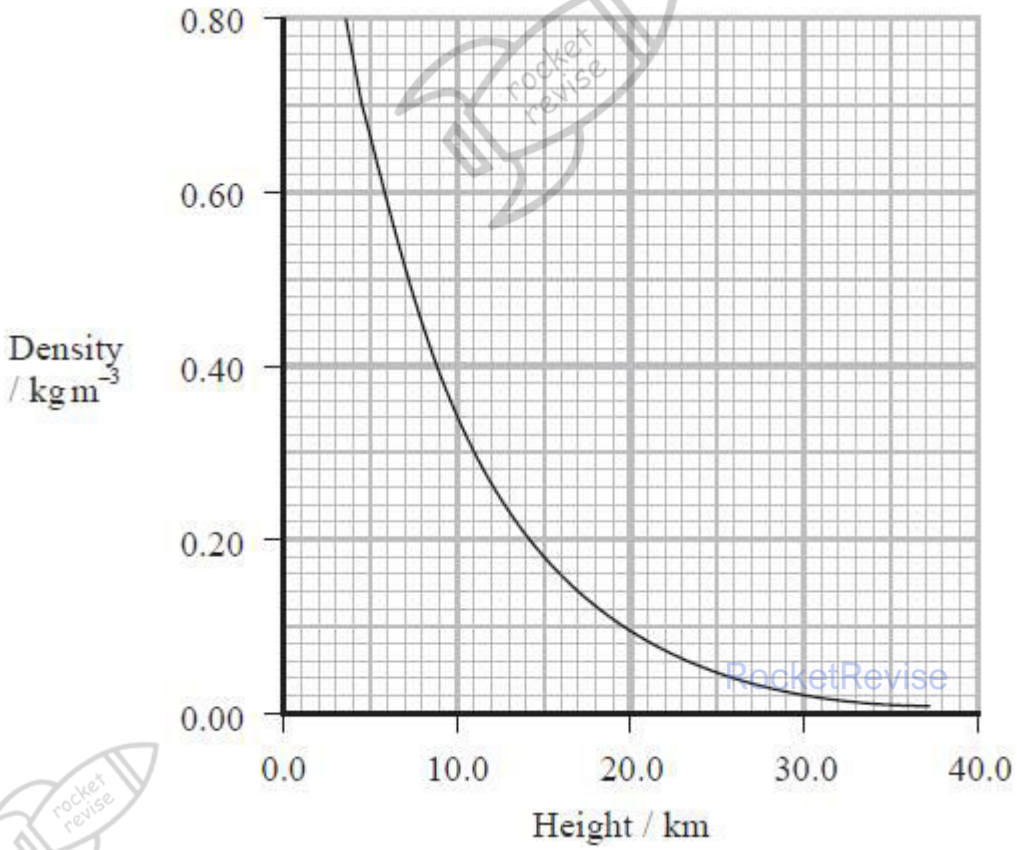
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(ii) The graph shows how the density of air decreases with height above the ground.




The scientists want to make measurements at a height of 25 km where the balloon would stop rising.
The balloon will expand as it rises and will burst if its volume exceeds 50 $m^3$.
Deduce whether the weather balloon could support the equipment at a height of 25 km, without bursting.

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*(c) A weather balloon made of a strong material will rise and eventually stop expanding. It will continue to rise until it reaches a maximum height.

Explain how and why the velocity of the weather balloon changes from release until it reaches its maximum height.
Ignore the effects of viscous drag force.

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**(Total for question = 14 marks)**

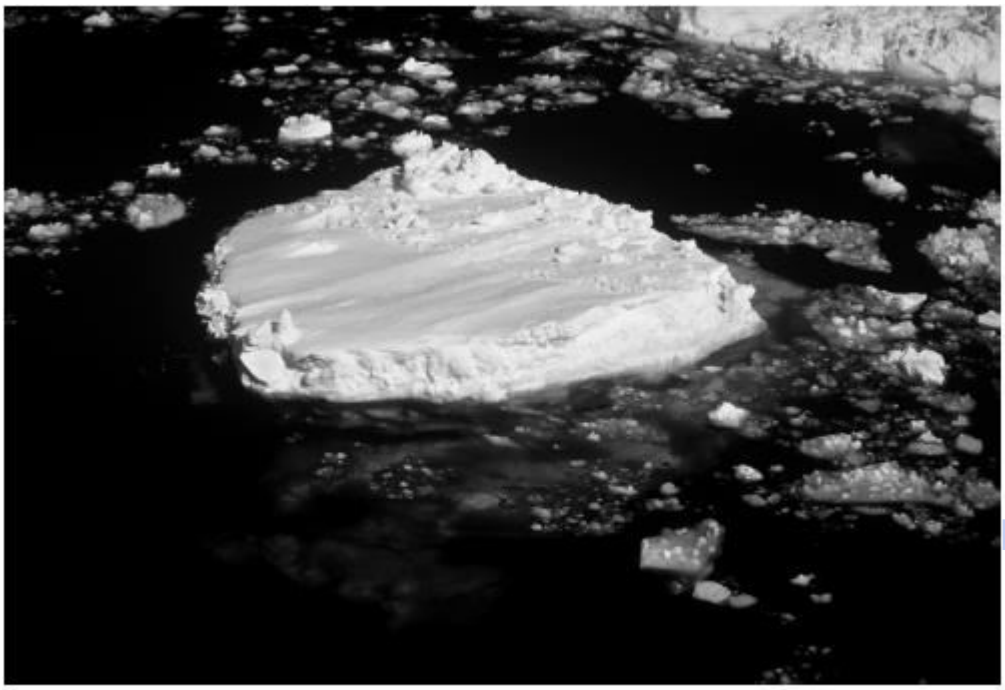
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**Q14.**

An iceberg is a large mass of ice that floats in seawater with most of its volume below the surface of the seawater. The photograph shows a typical iceberg.

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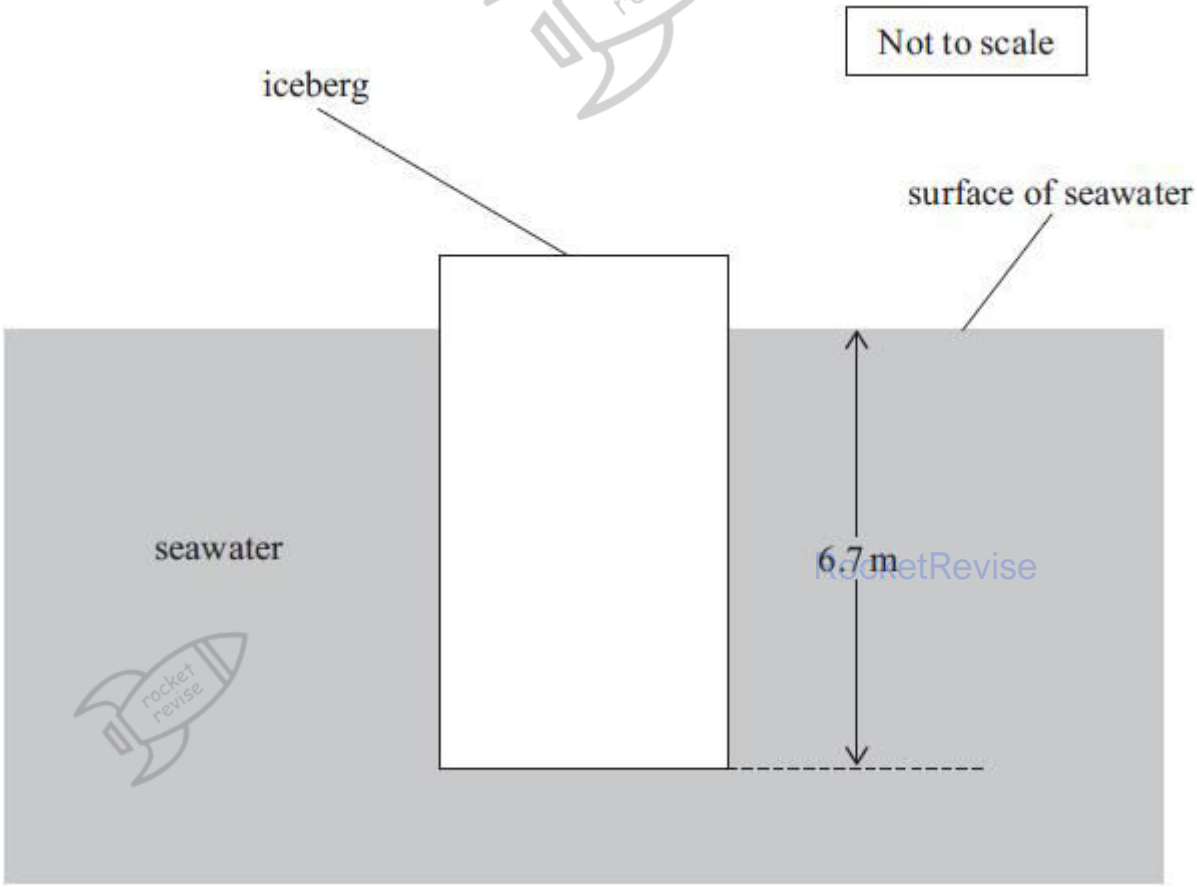
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(Source: © Doug Allan/Science Photo Library)

An iceberg can be modelled as a rectangular block of ice.

A particular iceberg has 6.7 m of its total height below the surface of the water, as shown.




The block of ice has a horizontal cross-sectional area of 3500 $m^2$.

(a) Calculate the weight of the block of ice.

density of seawater = $1.03 \times 10^3$ kg $m^{-3}$

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Weight of block = .....

(b) An identical block of ice floats in water of a lower density than the seawater in (a).

A larger volume of this block is below the surface of the water.
Explain why.

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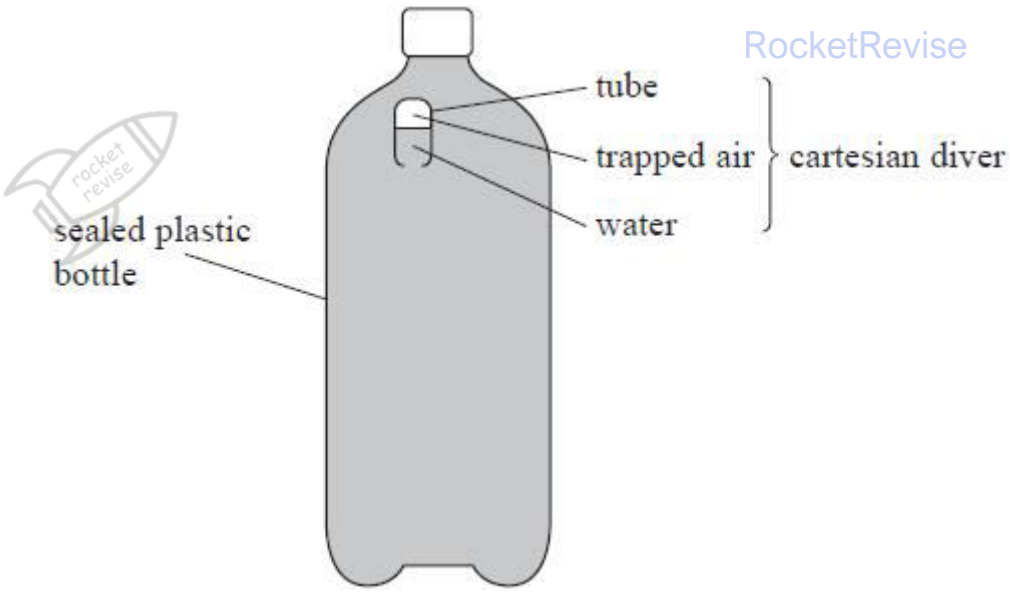
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**(Total for question = 5 marks)**

**Q15.**

A student has a toy called a 'cartesian diver' inside a sealed plastic bottle of water.

The cartesian diver consists of a tube closed at one end. Air is trapped by water inside the tube, as shown.




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The student squeezes the bottle. More water is forced into the tube, and the volume of the trapped air decreases.

Explain why the cartesian diver accelerates downwards.

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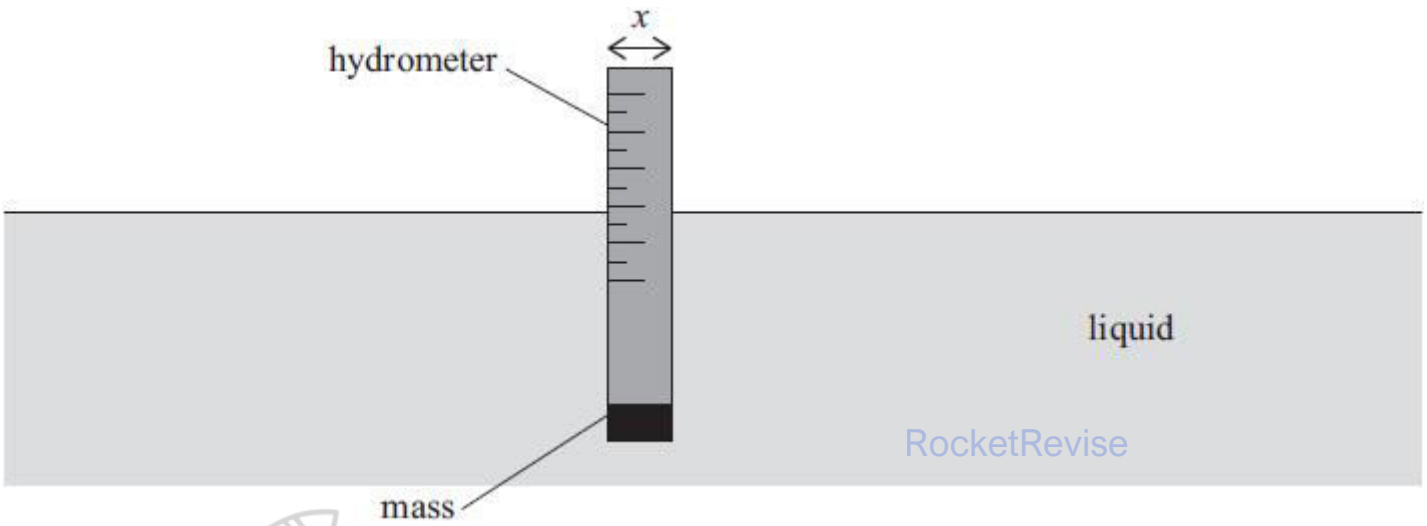
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**(Total for question = 3 marks)**

**Q16.**

A simple hydrometer is a cylinder with a mass at one end. The cylinder floats vertically in a liquid.

The diameter of the hydrometer is *x*. A scale on the side of the hydrometer shows the vertical position of the hydrometer in the liquid, as shown.




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A student placed the hydrometer in a sample of pure water.

(a) Complete the free-body diagram below to show the forces acting on a hydrometer when floating in water at equilibrium.

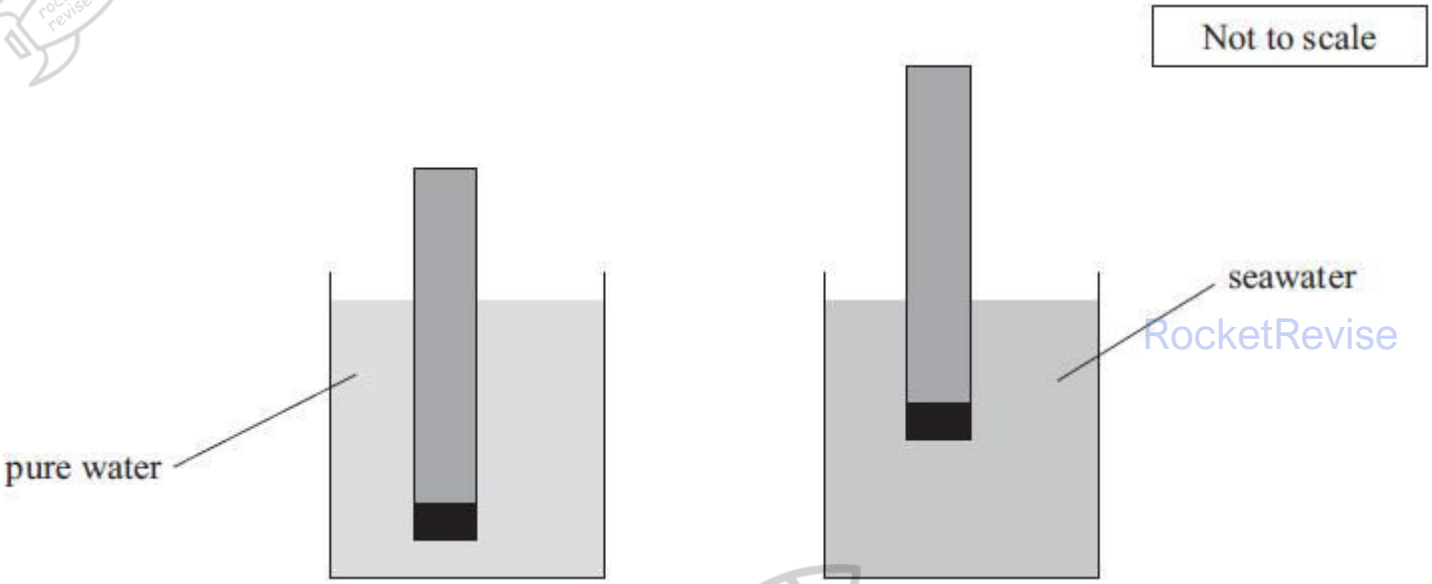
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(b) The student placed the hydrometer in a sample of seawater.

Seawater is more dense than pure water.
At equilibrium, the vertical position of the hydrometer was higher in the seawater than in the pure water, as shown.




(i) Explain why the vertical position of the hydrometer was higher in the seawater than in the pure water.

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(ii) Calculate the change in the vertical position of the hydrometer when moved from pure water into seawater.

weight of hydrometer = 0.324 N
density of pure water = 997 kg $m^{-3}$
density of seawater = 1025 kg $m^{-3}$
$x$ = 1.23 cm

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Change in vertical position = .....

**(Total for question = 10 marks)**

**Q17.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

A solid cube of weight $W$ is made from material with density $\rho$.

Which expression gives the length of each side of the cube?

☐ **A** $\left(\frac{\rho g}{W}\right)^3$

☐ **B** $\sqrt[3]{\frac{\rho g}{W}}$

☐ **C** $\left(\frac{W}{\rho g}\right)^3$

☐ **D** $\sqrt[3]{\frac{W}{\rho g}}$

**(Total for question = 1 mark)**

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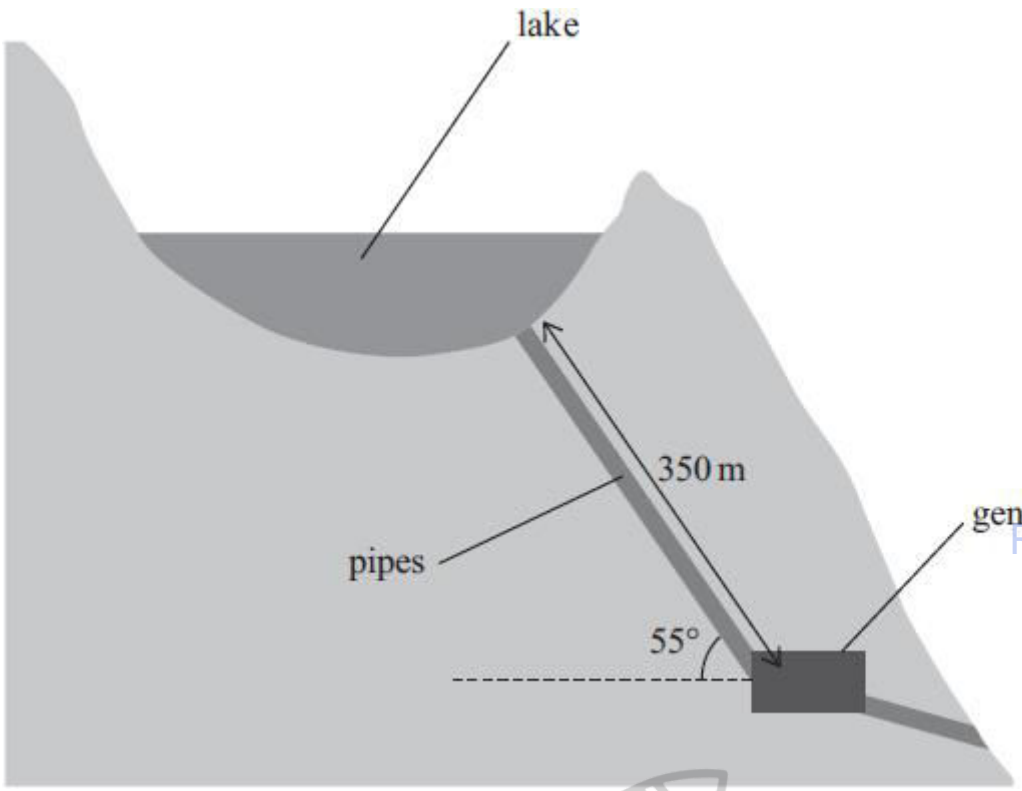
**Q18.**

The Cruachan hydroelectric power station in Scotland is built inside a mountain.

Water from a lake near the top of the mountain flows through straight pipes to a generator. The pipes have a length of 350 m and are inclined at an angle of 55° to the horizontal, as shown.

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The gravitational potential energy of the water in the lake is used to generate electrical power.

(a) In one minute, each pipe transfers 1700 $m^3$ of water to the generator. The density of the water in the lake is 1100 kg $m^{-3}$.

Show that the power transferred by the water in each pipe is about 88 MW.

**(5)**

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(b) Water travels from the lake to the generator through 4 identical pipes.

As water moves through the 4 pipes to the generator, the total power wasted is 256 MW.
Calculate the efficiency of the power station.

**(2)**

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Efficiency = .....

**(Total for question = 7 marks)**

**Q19.**

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**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

A cube of volume $V$ is placed in a liquid of density $\rho$. The cube displaces a weight $W$ of the liquid.

Which of the following gives the magnitude of the upthrust acting on the cube?

- ☐ **A** $V \times W$
- ☐ **B** $V$
- ☐ **C** $V \times \rho$
- ☐ **D** $W$

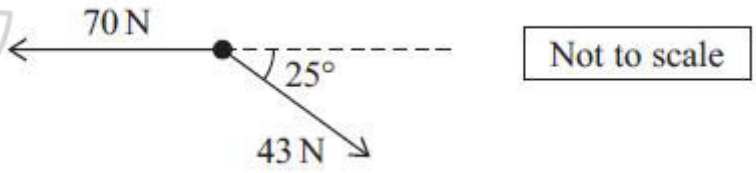
**(Total for question = 1 mark)**

**Q20.**

A small, remotely controlled submarine is used to explore the sea.

(a) The engine of the submarine produced a horizontal force of 70 N. The drag force on the submarine was 43 N at an angle of 25° to the horizontal, as shown.

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Determine the resultant of these two forces on the submarine by drawing a scaled vector diagram in the space below.

**(5)**

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Magnitude of resultant force = .....

Angle of resultant force to horizontal = .....

*(b) A crane on a ship can lift the submarine out of the sea, as shown.

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(Source: adapted from © PJF Military Collection/Alamy Stock Photo)

The submarine is initially completely underwater and at rest just below the surface.
A cable from the crane is attached to the submarine, so that the cable is vertical.
The crane pulls on the cable. The submarine moves vertically upwards at a constant speed until the whole submarine is out of the water.
Explain how the tension in the cable changes as the submarine moves from below the surface of the water to being completely out of the water.
You should assume the drag force on the submarine is negligible.

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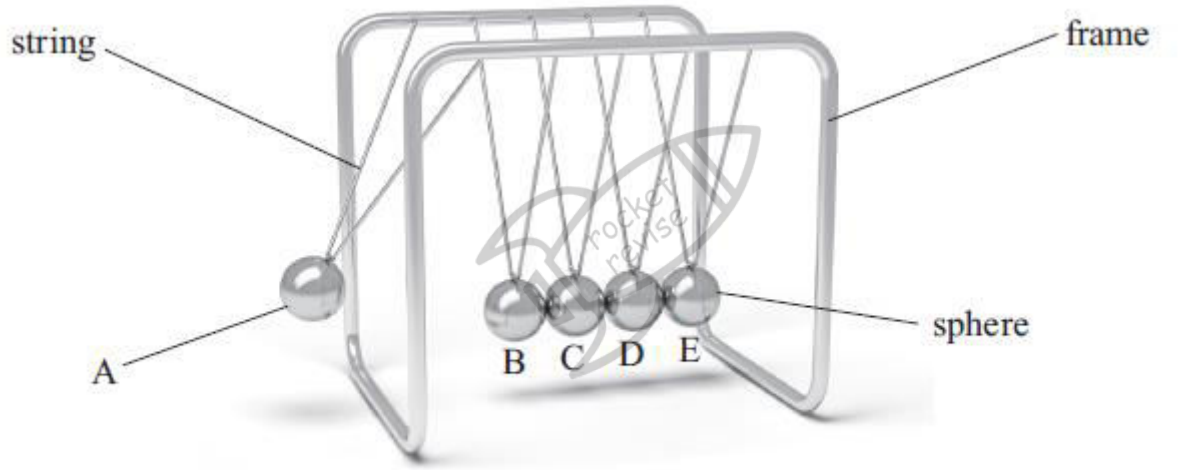
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**(Total for question = 11 marks)**

**Q21.**

A 'Newton's cradle' has five identical steel spheres, A, B, C, D and E. Each sphere hangs on string from a frame, as shown.

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(Source: © f9photos/Getty Images)

(a) Each sphere has a radius of $7.0 \times 10^{-3}$ m.

Calculate the mass of one sphere.
density of steel = $7.8 \times 10^{3}$ kg $m^{-3}$

**(3)**

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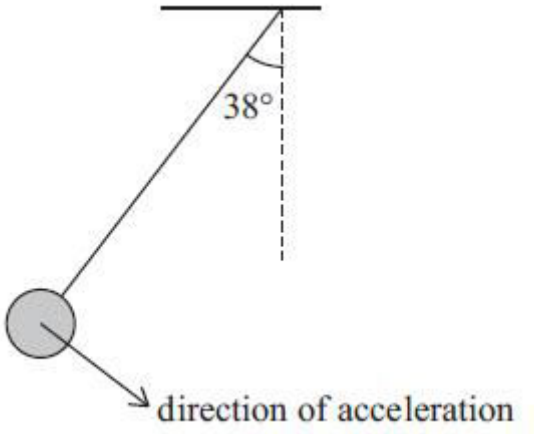
Mass of one sphere = .....

(b) A student pulls sphere A, so the string is at an angle of 38° to the vertical, as shown.

The student then releases sphere A.

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Determine the initial acceleration of sphere A when released.

(2)

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Initial acceleration = .....

*(c) Diagram 1 shows the position of the spheres when sphere A is released.

After being released, sphere A collides with sphere B and comes to rest.
Spheres B, C and D remain at rest. Sphere E moves away from sphere D.
Diagram 2 shows the position of the spheres when sphere E reaches the highest point.

Diagram 1

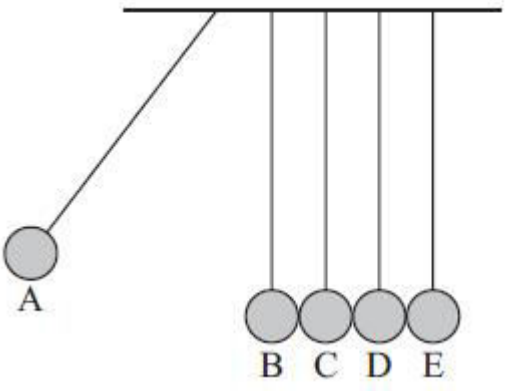
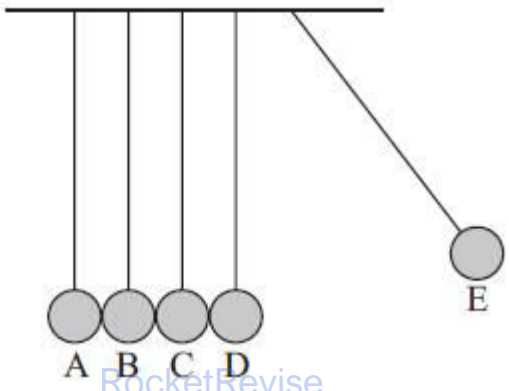



Diagram 2




Describe the changes in energy and momentum of spheres A and E, from the time sphere A is released until sphere E reaches its highest point.
Ignore the effects of air resistance.

(6)

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**(Total for question = 11 marks)**

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